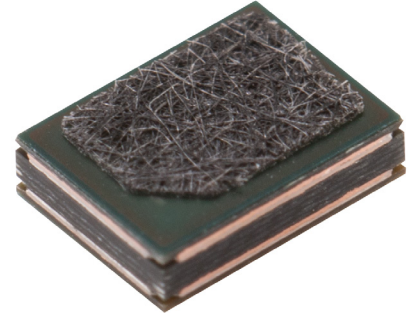


**MODEL:** CMM-3424DT-26165-TR | **DESCRIPTION:** MEMS MICROPHONE**FEATURES**

- IP67 rated
- top port
- digital
- omnidirectional

**ELECTRICAL**

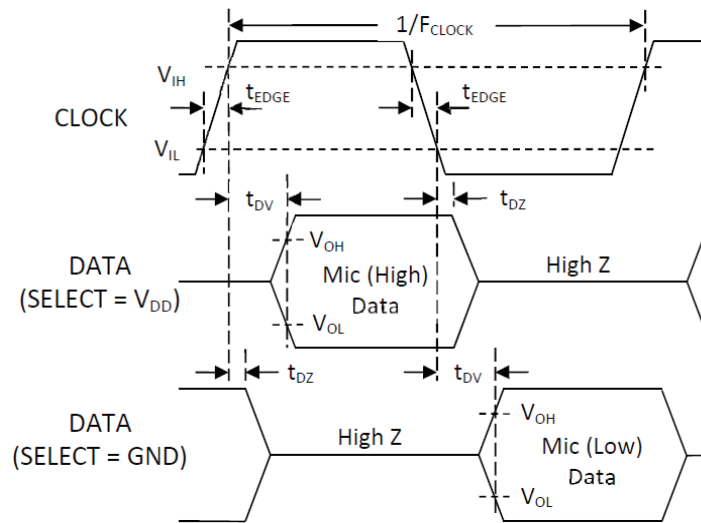
parameter	conditions/description	min	typ	max	units
directivity	omnidirectional				
sensitivity [S] ¹	at 94 dB SPL, 1 kHz	-27	-26	-25	dB FS
supply voltage [V _{DD}] ¹		1.6		3.6	V
current consumption [I _{DD}] ^{1, 6}	standard mode	600	800	1,000	μA
	low-power mode		300	400	μA
standby current [I _{STANDBY}] ^{5, 6}	F _{CLOCK} < 250 kHz		10	30	μA
signal to noise ratio [S/N]	at 94 dB SPL, 1 kHz [A-weighted]		65		dBA
total harmonic distortion [THD]	at 94 dB SPL, 1 kHz		0.15	0.3	%
	at 120 dB SPL, 1 kHz			10	%
acoustic overload point [AOP]	at 10% THD, 1 kHz		120		dB SPL
power supply rejection ratio [PSSR] [standard mode]	200 mVp-p sine wave @ 1 kHz V _{DD} =1.8 V		65		dB
	V _{DD} =3.3 V		65		dB
power supply rejection ratio [PSSR] [low-power mode]	200 mVp-p sine wave @ 1 kHz V _{DD} =1.8 V		72		dB
	V _{DD} =3.3 V		72		dB
power supply rejection [PSR] [standard mode]	100 mVp-p square wave @ 217 Hz V _{DD} =1.8 V		-80		dB FS
	V _{DD} =3.3 V		-80		dB FS
power supply rejection [PSR] [low-power mode]	100 mVp-p square wave @ 217 Hz V _{DD} =1.8 V		-90		dB FS
	V _{DD} =3.3 V		-90		dB FS
polarity	increasing sound pressure		increasing density of 1's		

DIGITAL INTERFACE

parameter	conditions/description	min	typ	max	units
clock frequency	standard mode	1		4.8	MHz
	low-power mode	350		800	kHz
fall-asleep time ^{2,3}	$F_{\text{CLOCK}} < 1 \text{ kHz}$, $V_{\text{DD}} = \text{On}$			10	ms
wake-up time ^{2,4}	$F_{\text{CLOCK}} \geq 1 \text{ MHz}$, $V_{\text{DD}} = \text{On}$			10	ms
data format	1/2 cycle PDM				
short circuit output current (I _{sc})	grounded output pin		1	10	mA
output load (C _{LOAD})				500	pF
logic input high (V _{IH})		0.65xV _{DD}			V
logic input low (V _{IL})				0.35xV _{DD}	V
logic output high (V _{OH})	I _{OUT} = 2 mA			V _{DD} +0.3	V
logic output low (V _{OL})	I _{OUT} = 2 mA	-0.3			V
clock duty cycle		40		60	%
clock rise time	t _{cr}			10	ns
clock fall time	t _{ct}			10	ns
delay time for valid data (t _{dv}) CLK=3.072 MHz oscilloscope	data trans high VDD=1.8 V	30		180	ns
	VDD=3.3 V	24		119	ns
	data trans low VDD=1.8 V	30		180	ns
	VDD=3.3 V	22		118	ns
delay time to for high Z (t _{dz}) CLK=3.072 MHz oscilloscope	data trans high VDD=1.8 V	1		14	ns
	VDD=3.3 V	1		10	ns
	data trans low VDD=1.8 V	1		14	ns
	VDD=3.3 V	1		10	ns
settling time (t _s)	VDD=1.8 V		13.5	25	ms
	VDD=3.3 V		13.5	25	ms
startup time	powered down => active mode				
	VDD=1.8 V		13.5	25	ms
	VDD=3.3 V		13.5	25	ms

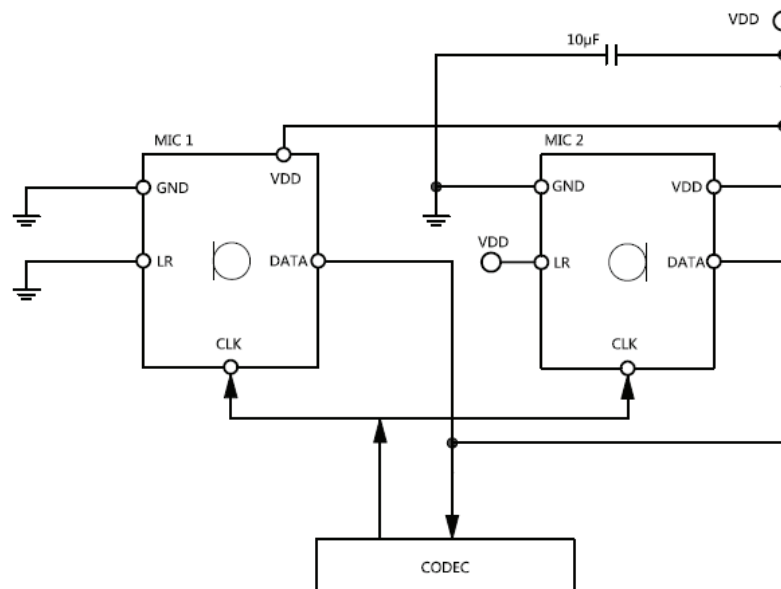
Notes: 1. 100% tested.
2. Valid microphone states are: Power Down Mode (mic off), Sleep Mode (low current, no output, fast start-up), and Active Mode (normal operation).
3. Time from fclock < 1kHz to sleep current specification is met when transitioning from Active to Sleep Mode.
4. Time from fclock ≥ 1MHz to all applicable specifications when transitioning from Sleep to Active Mode.
5. $\Delta I_{\text{DD}} = 0.5 \times V_{\text{DD}} \times C_{\text{LOAD}} \times f_{\text{clock}}$.
6. Specified max values are measured at VDD = +3.6V
7. All specifications measured at 15~25°C, humidity at 60~70%, VDD = 1.8 V, F_{CLOCK} = 3.072 MHz (768 kHz in low-power mode), no load, unless otherwise noted.

TIMING CHARACTERISTICS



Label	Select	Drives Data After	High-Z After
Data_H	High	rising clock edge	falling clock edge
Data_L	Low(default)	falling clock edge	rising clock edge

RECOMMENDED INTERFACE CIRCUIT



ENVIRONMENTAL

parameter	conditions/description	min	typ	max	units
operating temperature		-40		125	°C
storage temperature	in packaging	-40		125	°C
RoHS	yes				
IP level	IP67				

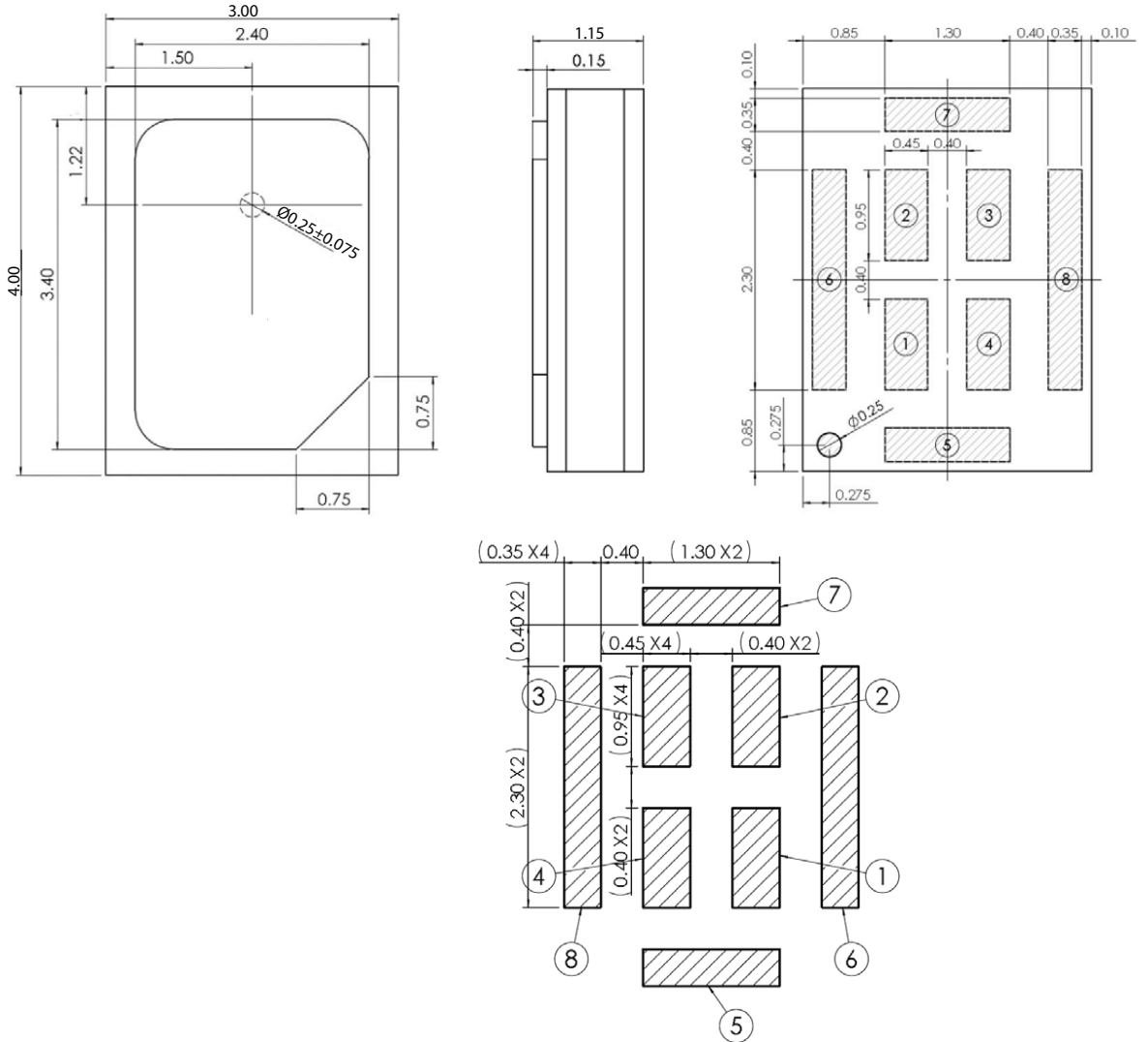
MECHANICAL

parameter	conditions/description	min	typ	max	units
dimensions	4.00 x 3.00 x 1.15				mm
acoustic port	top				
terminals	surface mount				
weight				0.03	g

MECHANICAL DRAWING

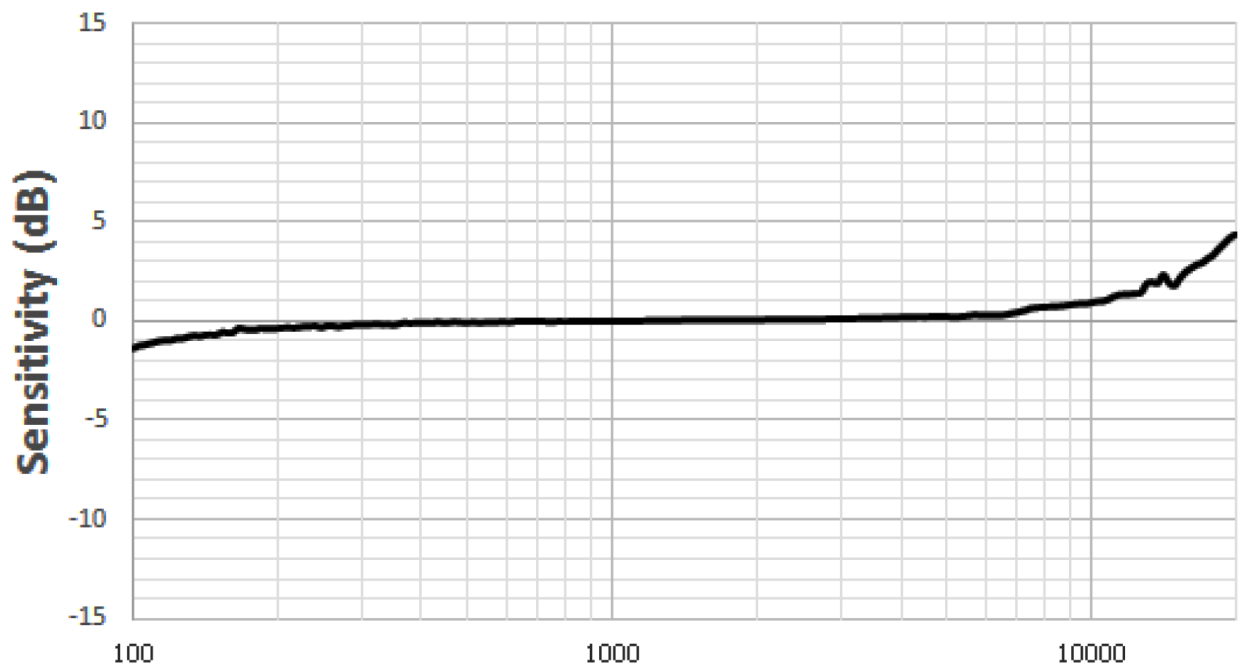
units: mm
tolerance: ±0.10 mm

TERMINAL CONNECTIONS	
TERM.	FUNCTION
1	VDD
2	select
3	CLOCK
4	DATA
5	GND
6	GND
7	GND
8	GND

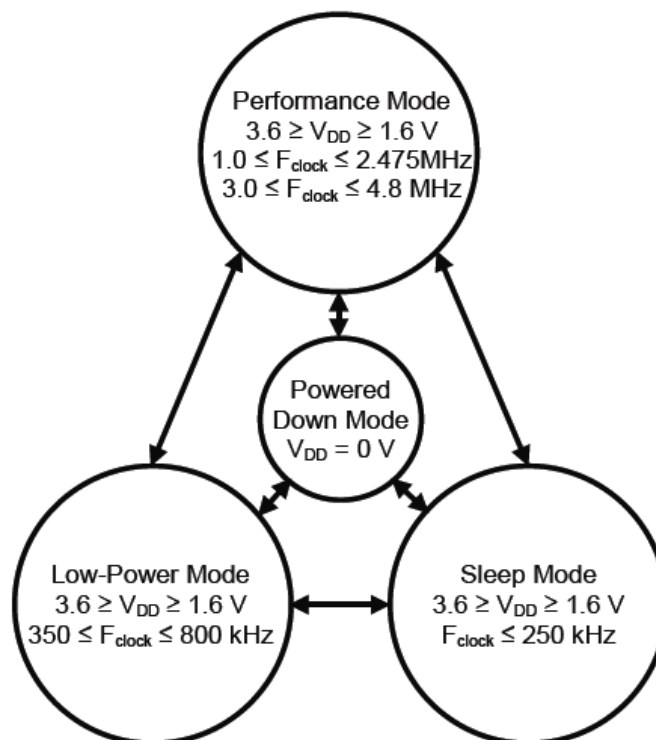


Recommended PCB Layout
Top View

FREQUENCY RESPONSE CURVE



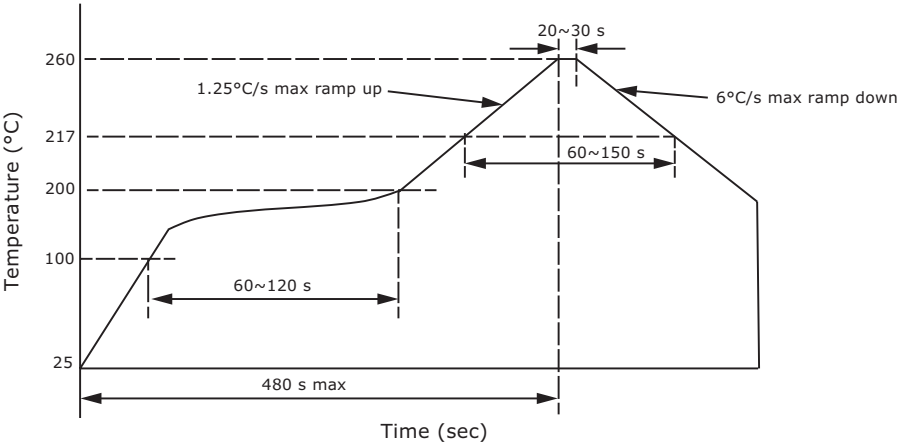
MICROPHONE STATE DIAGRAM



SOLDERABILITY

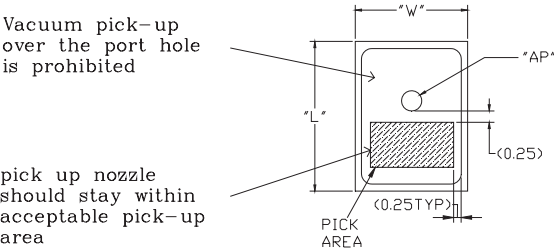
parameter	conditions/description	min	typ	max	units
reflow soldering ⁸	see reflow profile			260	°C

Note: 8. Not recommended to exceed 3 reflow cycles.



HANDLING RECOMMENDATIONS

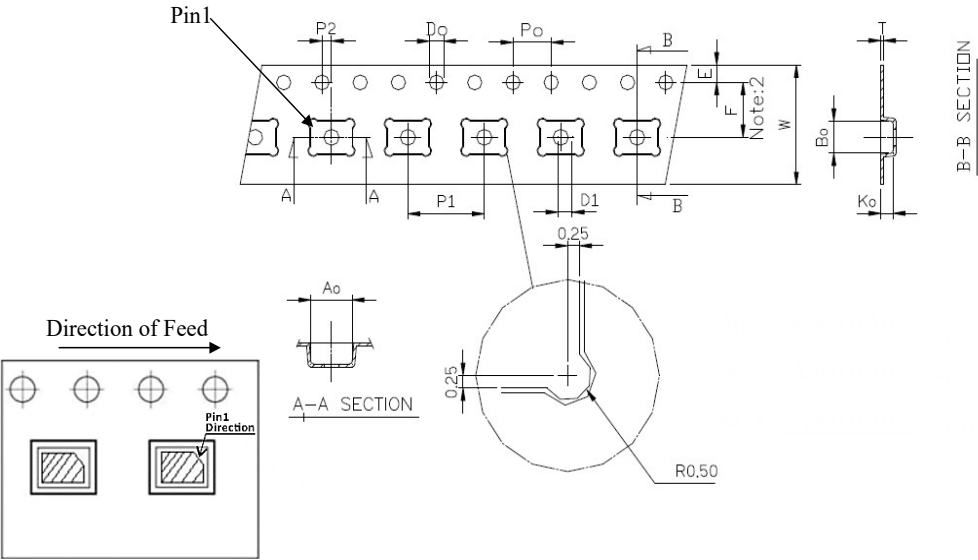
1. Not recommended to blow air heavily over acoustic port as debris could impact mic function.
2. Not suitable for wash process after reflow.
3. Not recommended to brush board with or without solvents after reflow process.
4. Not recommended to directly expose to ultrasonic processing or cleaning.
5. Not recommended to insert any object in port of device at any time.
6. Not recommended to apply over 30 psi of air pressure into the port hole.
7. Not recommended to pull a vacuum over port hole.
8. Not recommended to apply a vacuum when repackaging into sealed bag a rate faster than 0.5 atm/sec.
9. Not recommended to clean table or carried plate with air guarding system that could induce particle floating inside mic.



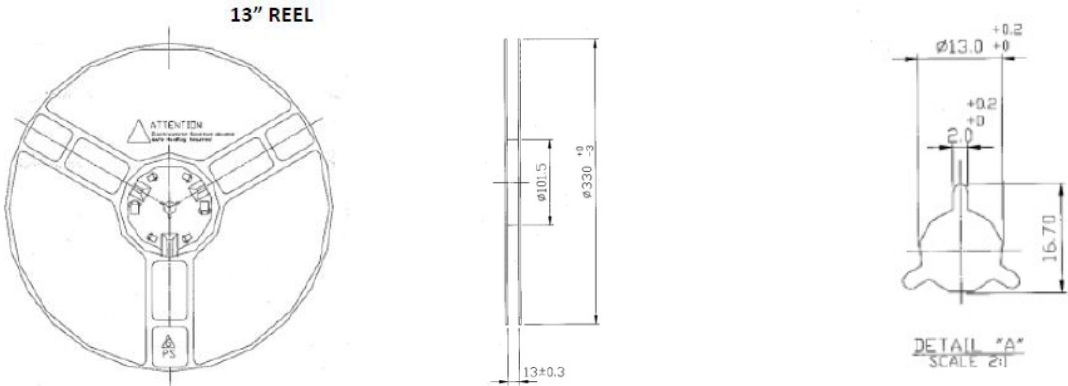
Recommended Vacuum Nozzle Pickup
Top View

PACKAGING

parameter	conditions/description	min	typ	max	units
MSL	Class 1				
reel size	Ø330 mm				
reel QTY	5,200 pcs per reel				
carton size	350 x 175 x 355 mm				
carton QTY	26,000 pcs				



Item	W	E	F	ØD0	D1	A0	K0
DIM(mm)	12.0±0.30	1.75±0.10	5.50±0.10	1.55±0.05	1.50 (min)	4.30±0.10	1.30±0.10
Item	P0	10P0	P1	P2	T	B0	--
DIM(mm)	4.0±0.10	40.0±0.10	8.0±0.10	1.0±0.10	0.30±0.05	3.20±0.10	--



REVISION HISTORY

rev.	description	date
1.0	initial release	05/17/2023
1.01	CUI Devices rebranded to Same Sky	09/11/2024
1.02	modified chip design circuit	02/27/2025

The revision history provided is for informational purposes only and is believed to be accurate.



Same Sky offers a one (1) year limited warranty. Complete warranty information is listed on our website.

Same Sky reserves the right to make changes to the product at any time without notice. Information provided by Same Sky is believed to be accurate and reliable. However, no responsibility is assumed by Same Sky for its use, nor for any infringements of patents or other rights of third parties which may result from its use.

Same Sky products are not authorized or warranted for use as critical components in equipment that requires an extremely high level of reliability. A critical component is any component of a life support device or system whose failure to perform can be reasonably expected to cause the failure of the life support device or system, or to affect its safety or effectiveness.

sameskydevices.com